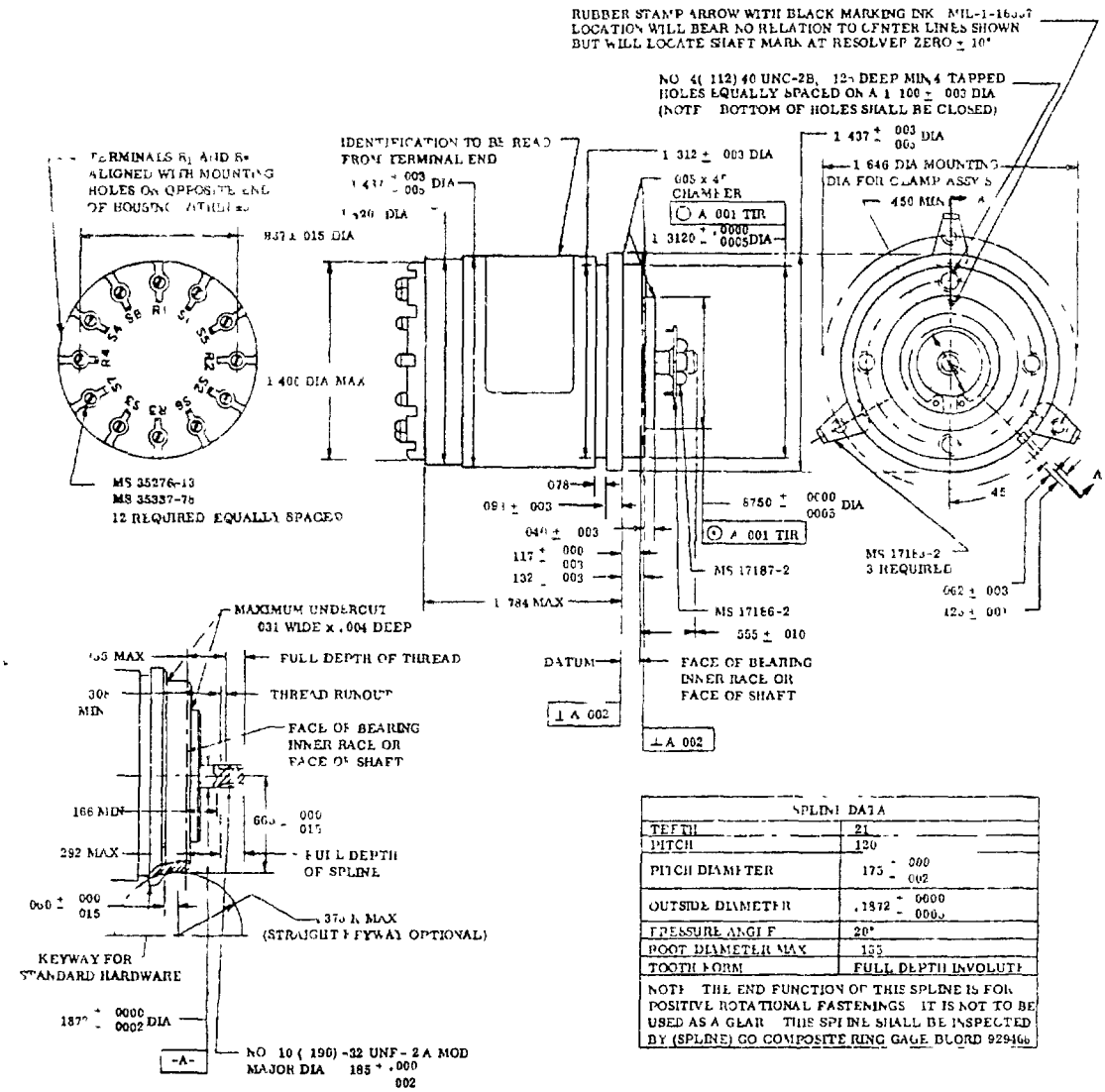
MILITARY SPECIFICATION SHEET
RESOLVER, ELECTRICAL MK4 MOD 0

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of Specification MIL-R-23417.



SPLINE DATA	
TEETH	21
PITCH	12/0
PITCH DIAMETER	1.75 +.000 -.002
OUTSIDE DIAMETER	.1872 +.0000 -.0005
PRESSURE ANGLE	20°
ROOT DIAMETER MAX	155
TOOTH FORM	FULL DEPTH INVOLUTE

NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. THIS SPLINE SHALL BE INSPECTED BY (SPLINE) GO COMPOSITE RING GAGE BUORD 929466

PARTIAL SECTION A-A

NOTES

1. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ± .005, FRACTIONS ± 1/64, ANGLES ± 1°
2. INTERPRET DIMENSIONING AND TOLERANCING IN ACCORDANCE WITH MIL-STD-6
3. ECCENTRICITY ON SMOOTH PORTION F OF SHAFT DIAMETER AT -A- DATUM SHALL NOT EXCEED .0008 TIR, WHEN SHAFT IS ROTATED IN BEARINGS
4. TERMINAL BLOCK AND END CAP MOUNTED ON RESOLVER INDEPENDENT OF TERMINAL SCREWS
5. TERMINAL LUGS, MS 17182-1 INSTALLED RADIALLY WITH BARRELS INBOARD OR OUTBOARD. RECESSES SHALL PREVENT LUGS FROM TURNING

FSC 5990

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TABLE I

Requirements:

Dimensions, Configuration, and Performance characteristics See Figure 1 and the following:

1-Excitation Winding	Stator
2-Frequency (CPS)	400
3-Input Voltage (VOLTS) rms	
Maximum	26
Test	15 ± 0.5
Minimum	5.0
4-Primary Current (MA) Max	15
5-Primary Power (WATTS) Max	0.05
6-D.C. Resistance (OHMS)	
Stator	165 ± 10%
Rotor	130 ± 10%
Compensating Winding, Resistor, or Both	215 ± 10%
7-Impedance (OHMS)	
Zro	200 ± 10% +J1200 ± 10%
Zso	230 ± 10% +J1050 ± 10%
Zrs	285 ± 10% +J125 ± 10%
Zco	275 ± 10% +J1100 ± 10%
8-Transformation Ratio	
Rotor/Stator	0.98 ± 0.01
9-Equality of Transformation Ratio (PERCENT) (MAX SPREAD)	0.2
10-Phase Shift (DEGREES) (Measured with ±.15 min accuracy)	
Nominal	8.5 ± 0.5
Due to Rotation	±0.5
Due to Voltage	±0.5
11-Interaxis Error (MINUTES) Max	
Rotor	±5
Stator	±7
12-Null Voltage (MV (rms)/V(rms) input)	
Fundamental	0.53
Total	1.00
13-Compensating Windings Fundamental Frequency Component of Null Voltage (MV (rms)/V(rms) input)	0.5
14-Equality of Compensating Windings (MV) Max	8.0
15-Function Error (PERCENT) Max	±0.1

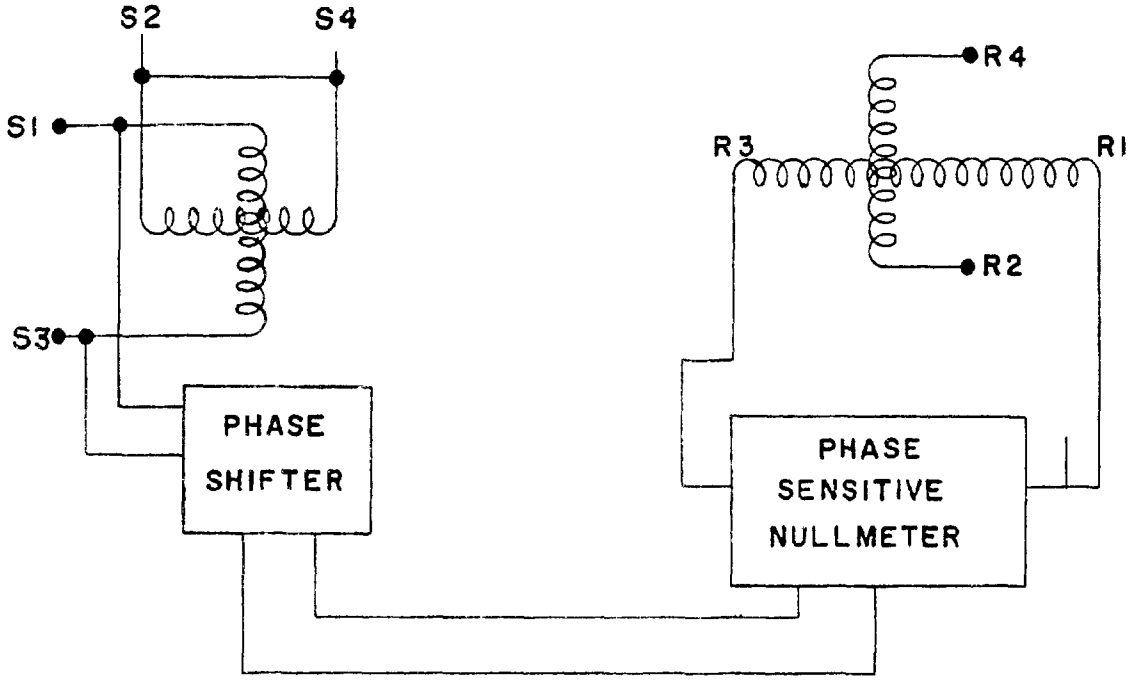
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TABLE I
(Continued)

16-Shift of Resolver Zero (MINUTES)		
With Voltage	±1	
With Frequency	±1	
17-Friction Torque (OZ-IN) Max	0.15	
18-Temperature Rise (DEGREES C) Max	20°	
19-Radial Play (INCHES) Max	.0006	on Reversal of 1/2 lb. Load
20-End Play (INCHES) Max	.0015	on Reversal of 1/2 lb Load

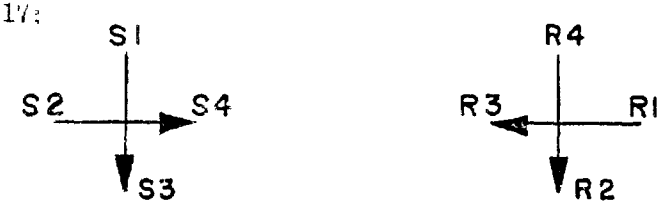
NOTES:

1. The following items are exceptions to MIL-R-23417 and are retained as specified in earlier documents to achieve interchangeability.
   a. Type designation - MK 4 Mod O
   b. Terminal identification (see dwg. page 1)
   c. Resolver zero - Use diagram below in lieu of Figure 2 in MIL-R-23417.



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a Phase rotation - Use information below in lieu of 3.4.2.2 in MIL-R-23417:

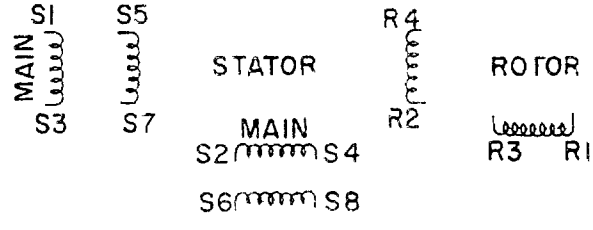


$E(R2-R4) = -KE\ (S1-S3)\cos\theta - KE\ (S2-S4)\sin\theta$
$E(R1-R3) = -KE\ (S2-S4)\cos\theta + KE\ (S1-S3)\sin\theta$

e. All diagrams, circuits, and tests applicable to stator excited resolvers in MIL-R-23417 may be utilized, providing that the MK 4 Mod 0 equivalent terminals are substituted for those designated, as follows:

MIL-R-23417 Designation	MK 4 Mod 0 Equivalent
S1	S1
S2	S2
S3	S3
S4	S4
R1	R4
R2	R3
R3	R2
R4	R1
C1	S5
C2	S6
C3	S7
C4	S8

2. A terminal identification diagram shall be added to the nameplate specified in MIL-R-23417:



Application for qualification shall be made to Chief, Bureau of Naval Weapons, Engineering Division, Components Branch Attn: RREN-4, Department of the Navy, Washington D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION
MIL-R-23417/6(WP) RESOLVER, ELECTRICAL: MK 4 MOD 0

ORGANIZATION (of submitter)	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426

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DEPARTMENT OF THE NAVY
Bureau of Naval Weapons
Washington, D. C 20360

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DEPARTMENT OF THE NAVY
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